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Estimation of Peanut Yield Losses Resulting From Early Leaf Spot on Spanish Cultivars in Oklahoma: A 26-Year Summary

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Abstract

Early leaf spot (ELS), caused by *Passalora arachidicola*, is the most prevalent and yield-limiting foliar disease affecting peanuts in Oklahoma. Quantifying yield losses associated with end-of-season defoliation caused by ELS is essential for evaluating the effectiveness of currently deployed control strategies. To that end, a meta-analysis was conducted to assess the heterogeneity in the relationship between ELS defoliation (percentage) and peanut yield (kilograms/hectare) of Spanish cultivars (*Arachis hypogaea* ssp. *fastigiata* var. *vulgaris*) in Oklahoma. Data were mined from fungicide efficacy trials performed in small plots across Oklahoma between 1990 and 2023. Studies (n = 49) over 26 years met the criteria of at least a 10% difference between the minimum and maximum defoliation within the study. A random coefficient model was successfully fitted to the data using maximum likelihood. The estimates of population average of the intercept and slope were $\widehat{\beta}_0 = 4{,}296.6$ kg/ha (SE = 131.9) and $\widehat{\beta}_1 = 13.7$ kg/ha (SE = 0.9), respectively. The damage coefficient, which represents the percentage reduction in yield per percentage point increase in crop defoliation, was 0.32%. These results allow the prediction of yield loss given for end-of-season defoliation levels and guide peanut producers and researchers in Oklahoma regarding modifying production practices and developing decision-making tools to mitigate losses resulting from ELS defoliation.

Keywords: crop losses, damage coefficient, meta-analysis, *Passalora arachidicola*

The United States ranks fourth in world peanut production, producing approximately 2.8 million metric tons on 586,928 ha in 2022 (USDA-FAS 2023; USDA-NASS 2023). In 2023, the eastern U.S. peanut-growing region (Virginia to Florida and west to Mississippi) supplied more than 87% of the country's crop (USDA-ERS 2023). Nearly all of the peanuts produced in this region are runner and Virginia peanut market types, which are both *Arachis hypogaea* ssp. *hypogaea* var. *hypogaea* (Stalker 2017; USDA-FSA 2023). In contrast, the southwest production region, consisting of Arkansas, Oklahoma, New Mexico, and Texas, produces all four major market types, including Spanish and Valencia (USDA-FSA 2023). Small- and red-seeded runner cultivars are sometimes also used to fill the Spanish and Valencia market, but taxonomically, Spanish and Valencia peanuts are distinct botanic varieties, *A. hypogaea* ssp. *fastigiata* var. *vulgaris* and *A. hypogaea* ssp. *fastigiata* var. *fastigiata*, respectively (Stalker 2017). Spanish and Valencia genotypes differ from runner and Virginia in that they have an upright canopy architecture, produce flowers from the main stem (Stalker 2017), mature earlier (Boote 1982; Marsalis et al. 2009), and generally yield less (Luper et al. 2007; Mixon and Branch 1985). Despite their lower yields, Spanish and Valencia are in demand for use in candy bars and all-natural peanut butter and for roasted, in-shell peanuts (Archer 2016; National Peanut Board 2023). Arid parts of the Southwest are also more conducive for organic peanut production than the Eastern United States, and Texas grows nearly all the organic peanuts produced in the United States (USDA-ERS 2023; Wann et al. 2011). Nonetheless, Southwestern producers share several diseases with growers in the East, including early leaf spot (ELS) caused by *Passalora arachidicola* (syn. *Cercospora arachidola* Hori) and late leaf spot (LLS) caused by *Nothopassalora personata* (syn. *Cercosporidium personatum* [Berk. & M.A. Curtis] Deighton) (York et al. 1994).

ELS and LLS are the most widespread of all peanut foliar diseases, and *P. arachidicola* and *N. personata* are among the most important diseases of peanut worldwide (Nutter and Shokes 1995; Shokes and Culbreath 1997). Managing crop rotation, irrigation, tillage, and planting dates have been shown to reduce leaf spot diseases, but fungicides are generally needed to protect yields regardless of cultural practices adopted (Branch et al. 2021; Brenneman et al. 1995; Damicone 2017; Fulmer et al. 2019; Monfort et al. 2004). Yield losses of up to 50% can result in untreated susceptible cultivars (Shokes and Culbreath 1997). Management of peanut diseases has been traditionally accomplished through applications of fungicides to reduce disease intensity and protect yields. Chlorothalonil, demethylation inhibitors, and strobilurin fungicides are effective for managing leaf spots if applied in a timely manner (Culbreath et al. 2008; Grichar et al. 2010), but their use can comprise a significant portion of input costs (Kemerait 2023; Kemerait et al. 2004). Estimated fungicide application costs per growing season can range from $100 to $300 per hectare, depending on the products used and the number of applications (Standish et al. 2019; Woodward et al. 2008).

In Oklahoma, ELS has been more prevalent than LLS for decades (Damicone 2017; Damicone et al. 1994; Melouk and Banks 1978; Nutter and Shokes 1995), perhaps partly because of the widespread planting of susceptible Spanish cultivars (Luper et al. 2007). Host plant resistance to ELS would offer effective disease control. Identification of genetic sources of resistance has resulted in the partially resistant cultivars being developed and released for production in the Southeastern United States (Gorbet et al. 1987;

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Gorbet and Tillman 2011; Holbrook and Culbreath 2007, 2008), but none have been developed thus far for the Southwest. As a result, Oklahoma peanut growers have suffered yield losses owing to severe crop defoliation caused by leaf spots in disease-conducive years. Hours of relative humidity ≥95% and temperatures between 16 and 30°C are known to favor ELS infection and spread in the field (Damicone et al. 1994; Jewell 1987). In addition, virtually all (>99%) of Oklahoma's peanut acreage is irrigated, in contrast to the Eastern United States, adding a different component to the production system and possibly increasing wetness periods (Damicone et al. 1994; Porter and Wright 1991).

Recently, Anco et al. (2020) estimated yield losses caused by LLS and ELS defoliation for Virginia and runner peanuts grown mostly in the Eastern United States. That study estimated defoliation thresholds for yield loss of Virginia and runner types at 40 and 50%, respectively. A similar study has yet to be conducted for the Spanish cultivars in the irrigated and arid southwest production region. In addition, the economic damage threshold (Zadoks 1985), defined as the level of crop damage at which the benefit of control exceeds its costs, has also yet to be established. Therefore, we gathered ELS defoliation and peanut yield data from fungicide trials conducted over 33 years (1990 to 2023) in Oklahoma. Our objectives were to explore and summarize the relationship between peanut yield and ELS defoliation in Spanish market type peanuts. This information is essential for estimating losses in yield based on end-of-season defoliation levels at the field level resulting from ELS and to assist farmers with making informed decisions about disease management.

Materials and Methods

Data source and criteria for inclusion of studies

Data were mined from fungicide efficacy trials performed in small plots across Oklahoma between 1990 and 2023. The data were mostly from nonrefereed trial reports published by the Oklahoma Agricultural Experiment Station. A portion of this data has been used in a brief report assessing the relationship between peanut yield losses and foliar and soilborne diseases in Oklahoma (Damicone et al. 2018). Field plots consisted of four 7.62-m-long rows spaced 0.9 m apart. The commercially available Spanish cultivars Spanco (Kirby et al. 1989), Tamspan 90 (Smith et al. 1991), and OLé (Chamberlin et al. 2015) were used to conduct the experiments. The experimental design was a randomized complete block with four replications, each separated by a 1.5-m-wide fallow buffer. All data were taken from the two center rows. Trials were conducted following approved recommendations for agronomic practices for peanuts in the state (Godsey et al. 2011). Field trials were generally planted in May, grown under sprinkler irrigation, and harvested in October. Fungicide programs typically consisted of either 14-day calendar programs totaling six applications per season or weather-based programs totaling three to five applications per year (Damicone et al. 1994). Fungicide programs were initiated 45 days after planting at the R2 to R3 peanut growth stage, when the plant canopy favors leaf spot development. Fungicides were applied broadcast through flat-fan nozzles (8002VK) spaced 0.46 m apart with a CO_2-pressurized wheelbarrow sprayer at 18 to 23 liters/ha. The specific fungicide treatments and respective application rates were variable among trials. All studies were conducted under natural inoculum levels and included an untreated control treatment.

The disease response variable used in our study was ELS defoliation (percentage), representing the visual assessment of the percentage of leaflets dropped in each plot caused by ELS approximately 15 to 30 days after the last fungicide application and/or just before digging (at approximately the R6 growth stage; full seed/< 21 days before peanut inversion) (Boote 1982). The plots were dug and inverted and windrowed, and the two central rows of each plot were harvested with a combine. The pods were dried to 9% moisture and cleaned before weighing.

Only studies reporting peanut yield and ELS defoliation (percentage) in the same trial were included in the dataset. To be included in the analysis, trials needed to have within-study differences between the minimum and maximum defoliation of at least 10 percentage points (Barro et al. 2023; Lehner et al. 2017; Madden and Paul 2009). To estimate the yield losses caused by ELS in Spanish peanuts in Oklahoma, data from 49 trials over 26 years were used to model the yield-ELS defoliation relationship (Table 1).

Effect sizes and meta-analytic modeling

Intercepts (β_0) and slopes (β_1) (using ordinary least square regression modeling) were estimated for the linear relationship

Table 1. Summary information for 49 fungicide trials conducted in Oklahoma from 1990 to 2023 to evaluate the efficacy of fungicides on early leaf spot (ELS) defoliation (percentage) and peanut yield (kilograms/hectare)

				Range[e]	
Study[a]	*N*[b]	Year[c]	Market type[d]	ELS defoliation (%)	Peanut yield
1	5	1990	Spanish	8.2–91.5	2684–3781
2	3	1991	Spanish	0.0–13.5	3273–3699
3	5	1991	Spanish	3.5–49.0	3435–4066
4	4	1991	Spanish	16.9–92.5	2066–3117
5	5	1991	Spanish	0.0–75.6	2988–4846
6	9	1991	Spanish	0.0–29.4	4126–4879
7	10	1992	Spanish	4.7–92.7	3649–5588
8	7	1992	Spanish	6.0–90.0	2252–4218
9	6	1993	Spanish	1.0–76.0	1725–2354
10	10	1993	Spanish	4.4–76.1	3562–4594
11	10	1994	Spanish	34.6–96.7	3110–4919
12	12	1994	Spanish	2.5–76.4	2806–3882
13	13	1995	Spanish	0.3–37.5	1748–2302
14	12	1995	Spanish	0.0–30.0	1781–2228
15	10	1995	Spanish	1.0 68.8	3394–4238
16	14	1996	Spanish	3.0–86.0	2073–3619
17	16	1997	Spanish	1.0–91.3	2269–4276
18	12	1997	Spanish	0.0–78.8	2276–3761
19	13	1997	Spanish	1.0–87.5	2098–3424
20	10	1998	Spanish	1.0–77.5	3871–5887
21	12	1998	Spanish	0.0–61.0	3789–4642
22	19	2001	Spanish	0.0–63.8	3220–4456
23	12	2003	Spanish	0.0–77.0	2838–3155
24	16	2005	Spanish	23.3–97.5	2390–4212
25	15	2007	Spanish	0.0–62.1	2622–3405
26	10	2008	Spanish	0.0–57.5	3228–4220
27	10	2008	Spanish	0.0–74.6	3627–4627
28	10	2009	Spanish	0.0–70.0	3984–4749
29	10	2009	Spanish	0.0–66.6	4220–4903
30	10	2010	Spanish	6.6–42.9	4277–5041
31	10	2010	Spanish	1.7–71.2	3138–5350
32	10	2013	Spanish	0.4–82.1	2521–4098
33	10	2014	Spanish	18.7–71.7	3497–5529
34	10	2015	Spanish	0.0–65.0	2561–3936
35	10	2016	Spanish	0.0–97.5	2813–5009
36	10	2016	Spanish	0.0–85.0	4106–5741
37	10	2017	Spanish	10.0–100.0	2895–4700
38	10	2018	Spanish	17.4–100.0	2716–6131
39	4	2018	Spanish	0.0–100.0	1962–4283
40	10	2019	Spanish	0.0–47.5	3821–4854
41	10	2019	Spanish	0.0–47.5	3008–4375
42	13	2019	Spanish	0.0–72.5	2976–4268
43	12	2019	Spanish	0.0–76.6	3041–4432
44	11	2020	Spanish	0.0–28.0	4098–6033
45	6	2021	Spanish	10.0–75.0	3214–4603
46	6	2021	Spanish	8.0–75.0	2901–4077
47	9	2023	Spanish	1.4–31.2	5554–6285
48	4	2023	Spanish	2.7–31.2	5554–6066
49	7	2023	Spanish	1.2–31.2	5554–6098

[a] Study number.
[b] Number of observations within each study (*N*).
[c] Year the experiment was conducted.
[d] Peanut market type.
[e] Minimum and maximum values of ELS defoliation (percentage) and peanut yield (kilograms/hectare) per trial.

between ELS defoliation and peanut yield at the trial level (Dalla Lana et al. 2015; Duffeck et al. 2020; Madden and Paul 2009). In this case, β_0 represents the expected yield (kg ha^{-1}) when ELS defoliation is 0 and β_1 is the decline in expected yield with a unit increase in ELS defoliation (kg ha^{-1} %$^{-1}$), wherein a negative β_1 indicates that peanut yield declines with increasing ELS defoliation. The distribution of the linear coefficients estimated independently for each trial was summarized by calculating the interdecile (ID) range (90% − 10%), or 80% of the estimated intercept and slope values (Lehner et al. 2017; Madden and Paul 2009).

A random coefficient model was also fitted to the yield-ELS defoliation data to account for the random effect of each study on the intercept and slope. The population-average intercept and slope were estimated, assuming a linear relationship between ELS defoliation and yield (Lehner et al. 2017; Madden and Paul 2009). The predicted study-specific intercepts ($\widehat{\beta}_0$) and slopes ($\widehat{\beta}_1$), also known statistically as the best linear unbiased predictors (BLUPs) of the study effect, were estimated using maximum likelihood together with the population-average parameters. The *lmer* function of the *lme4* package of R (Bates et al. 2015) was used to estimate the mean effect based on the between-study variance and within-study variance and also predict the study-specific intercept ($\widehat{\beta}_0$) and slope ($\widehat{\beta}_1$) coefficients, as performed elsewhere (Barro et al. 2023; Duffeck et al. 2020; Lehner et al. 2017).

Relative yield loss estimation

The damage coefficient (*Dc*), expressed as the percentage relative yield loss, was calculated using the mean estimates of $\widehat{\beta}_0$ and $\widehat{\beta}_1$ obtained by the random coefficient model. The *Dc* was calculated by dividing the estimated population-average slope ($\widehat{\beta}_1$) (kilograms per hectare per percent) by the estimated population-average intercept ($\widehat{\beta}_0$) (kilograms per hectare) times 100 (Dalla Lana et al. 2015; Madden and Paul 2009). In this study, *Dc* represents the percentage of yield reduction per percentage increase in ELS defoliation.

Results

Yield-severity relationship, simple linear regression model

Across the 49 selected studies, ELS defoliation ranged from 0 to 100% (mean = 21.1%), wherein peanut yield ranged from 1,725 to 6,285 kg/ha (mean = 3,936 kg/ha) (Fig. 1A and B). The fit of the simple linear regression model at the study level resulted in a wide range of estimated intercepts and slopes (Fig. 1C and D), and it showed that peanut yield decreased as ELS defoliation increased. The estimated intercepts (β_0), which represent the attainable yields when ELS defoliation is absent, ranged from 2,009 to 6,366 kg/ha (mean = 4,308 kg/ha) and had an ID of 2,432 kg/ha (Fig. 1C; Table 2). The estimated slopes (β_1) were negative, ranging from −46.6 to −0.74 kg/ha (mean = −14.1 kg/ha) with an ID of −17.3 kg/ha (Fig. 1D). All slopes (β_1) were negative, indicating that peanut yield decreased as ELS defoliation increased (Fig. 2A).

Random coefficient model

The mixed model was successfully fitted to the data across the 49 trials using maximum likelihood. Mixed model–derived intercept and slope values varied among individual studies (Table 2). The estimated BLUPs for the intercepts ($\widehat{\beta}_0$) ranged from 2,018 to 6,117 kg/ha (ID = 2,330.7 kg/ha), and study-specific slopes ($\widehat{\beta}_1$) ranged from −23.0 to −3.9 kg/ha (ID = −10.3). The estimates of population average of the intercept and slope were $\widehat{\beta}_0$ = 4,296.6

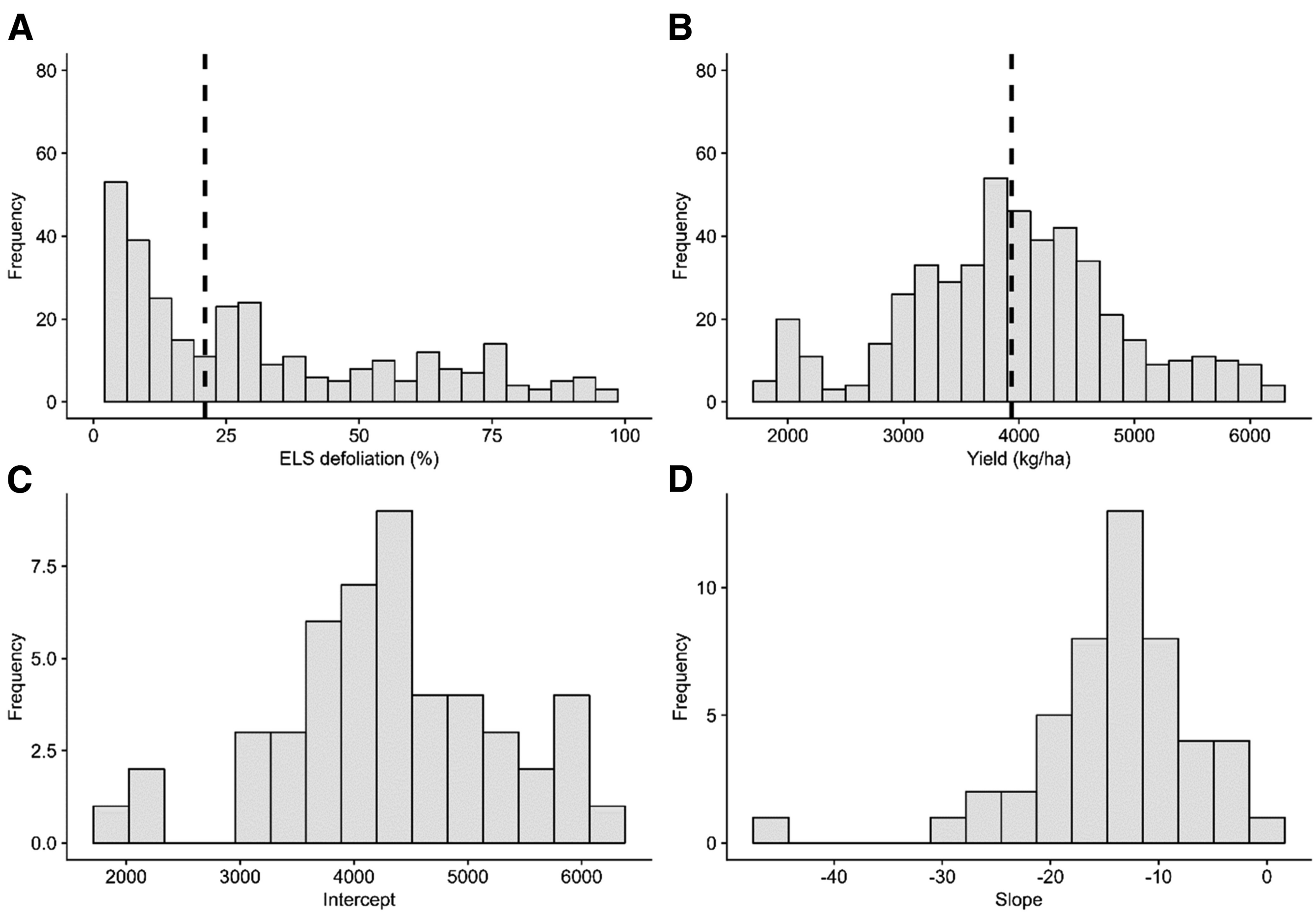

Fig. 1. The distribution of early leaf spot (ELS) defoliation (percentage) (**A**) and peanut yield (**B**) for the 49 selected studies, with the vertical dashed thick lines representing the mean of the respective variable. The distribution of estimated intercepts (**C**) and slopes (**D**) estimated by simple linear regression.

kg/ha (SE = 131.9) and $\widehat{\beta}_1$ = 13.7 kg/ha%$^{-1}$ (SE = 0.9), respectively (Fig. 2B). Both parameters were statistically different from 0 (P < 0.001).

The value of ELS defoliation (percentage), where a certain yield or yield loss is expected to occur, was also estimated based on Equation 12, presented by Madden and Paul (2009). For this study, the predicted disease defoliation (percentage) at which the absolute yield is reduced by 1,000 kg/ha would occur at an ELS defoliation of 73.0%.

Table 2. Coefficients estimated by a simple linear regression model fitted to data at the study level and estimated coefficients best linear unbiased prediction [EBLUP]) by a random coefficients model fitted to data on the relationship between peanut yield (kilograms/hectare) and early leaf spot defoliation (percentage) in 49 field trials conducted in Oklahoma from 1990 to 2023, with variable number of observations (N)

		Linear regression model coefficients (estimates)		Random coefficient model predictions (EBLUPs)	
Study code	N	β_1	β_0	$\widehat{\beta_1}$	$\widehat{\beta_0}$
1	5	3617.43	−7.68	3731.31	−10.49
2	3	3557.12	−21.07	3537.25	−10.44
3	5	3907.68	−9.65	3933.24	−11.88
4	4	3441.98	−14.66	3305.04	−10.54
5	5	4350.42	−18.15	4298.75	−14.59
6	9	4647.15	−8.01	4742.36	−15.10
7	10	5464.80	−19.19	5426.49	−18.69
8	7	4525.31	−19.60	4371.27	−15.47
9	6	2075.50	−1.73	2169.00	−4.09
10	10	4865.42	−15.68	4869.93	−15.94
11	10	5936.30	−27.74	5479.14	−19.94
12	12	3771.99	−13.38	3741.91	−11.77
13	13	2009.27	−5.41	2017.94	−3.92
14	12	2064.45	−4.35	2082.22	−3.99
15	10	3973.94	−8.85	4011.40	−11.68
16	14	3565.96	−13.54	3515.35	−11.31
17	16	4143.08	−16.86	4083.80	−14.43
18	12	3691.19	−15.46	3635.44	−12.12
19	13	3149.66	−12.05	3128.11	−9.62
20	10	5449.25	−17.95	5442.58	−18.51
21	12	4489.90	−11.45	4503.54	−14.12
22	19	3809.02	−9.22	3832.68	−10.99
23	12	3002.57	−0.75	3083.38	−6.48
24	16	4456.24	−13.94	4472.43	−14.26
25	15	3016.91	−2.32	3045.54	−7.24
26	10	3960.05	−12.73	3961.62	−12.31
27	10	4243.24	−8.28	4274.61	−12.46
28	10	4356.30	−5.32	4390.47	−12.36
29	10	4547.06	−4.02	4625.05	−13.20
30	10	4736.04	−9.42	4820.60	−15.58
31	10	4948.39	−25.64	4823.27	−17.70
32	10	3791.81	−14.16	3750.00	−12.03
33	10	5381.93	−22.18	5263.06	−18.34
34	10	3783.73	−18.81	3754.10	−12.62
35	10	5077.60	−16.88	5070.10	−16.94
36	10	5259.904	−13.66	5271.80	−16.80
37	10	4867.47	−10.17	5074.67	−14.05
38	10	6366.227	−28.23	6116.73	−23.00
39	4	4037.77	−11.30	4088.79	−12.32
40	10	4471.40	−13.93	4471.77	−14.41
41	10	4105.77	−23.60	4059.64	−13.76
42	13	3948.02	−13.24	3944.70	−12.40
43	12	4215.67	−15.52	4202.15	−13.79
44	11	5402.32	−46.58	5331.25	−19.33
45	6	4400.53	−12.80	4429.27	−13.97
46	6	4225.67	−12.66	4250.57	−13.37
47	9	6045.59	−14.96	6065.42	−21.18
48	4	6031.97	−15.20	6042.16	−21.11
49	7	5917.55	−11.87	5990.91	−20.79

Model-predicted yield losses

Damage coefficient (Dc) was calculated to consider yield or yield losses on a relative scale. Based on the population-average intercept (4,296.6 kg/ha) and slope (−13.7 kg/ha) estimated by the random coefficients model, the rate of decline in relative yield with increasing ELS defoliation (percentage) was estimated to be 0.32% (95% confidence interval = 0.26 to 0.38). Based on data from fungicide efficacy trials conducted with Spanish peanut cultivars in Oklahoma, a 1% increase in ELS defoliation at field conditions would result in a 0.32% peanut yield reduction.

Discussion

Our results confirm that ELS is a yield-limiting disease of Spanish cultivars in Oklahoma. A significant negative slope obtained through the random coefficient model confirmed the negative linear relationship between ELS defoliation and peanut yield. As expected, peanut yield decreased as ELS defoliation increased. However, there is significant variation in aspects of the relationship among studies, especially regarding the height of the regression lines (Fig. 2A). From our study, we calculated that for each unit increase in ELS defoliation, yield declined by 0.32 percentage points, or, in other words, the average reduction of the attainable yield is expected to be 32% when there is maximum observed defoliation (ELS defoliation = 100%). One possible explanation may be the shorter time required for the Spanish cultivars to reach maturity. The maximum crop defoliation occurs close to harvest time, minimizing the impact on yield losses. An example of using this relationship is that a peanut field with an expected yield of 4,400 kg/ha (intercept) and an ELS defoliation of 1% could experience a yield reduction of approximately 14 kg/ha.

A similar damage coefficient observed in our study (0.32%) was obtained by Anco et al. (2020). In that study, conducted with Virginia peanut cultivars in Virginia, North Carolina, and South Carolina, 12 kg/ha (or 0.27%) of peanut yield losses were observed per each 1% defoliation at 25% defoliation. However, the rate of change was much higher at 95% defoliation, with 38 kg/ha (0.87%) of yield losses per 1% defoliation. In that study, the nonlinear rate of change increased based on the increased defoliation level experienced by the crop, with higher yield losses expected at higher defoliation levels by Virginia market type.

For runner peanuts, higher yield loss rates, 57 kg/ha yield losses per unit defoliation increase, were reported in Alabama with the cultivar Florunner compared with our results (Backman and Crawford 1984). Yield losses varying from 33 to 99 kg/ha among individual location years were also reported for 'Florunner' in Georgia by Nutter and Littrell (1996). Anco et al. (2020), using data from Alabama, South Carolina, Florida, and Texas, obtained the most similar results to our study, where the runner market type yield loss rate was 11 kg/ha per unit defoliation, only three percentage points lower than the Spanish market type (14 kg/ha) in this study.

The damage coefficient (Dc) estimated from Spanish peanut yield and ELS defoliation (0.32%) is notably lower relative to Dc calculated for foliar diseases in other crops, such as frogeye leaf spot of soybean caused by *Cercospora sojina* (0.51%) (Barro et al. 2023), target spot caused by *Corynespora cassiicola* (0.48%) (Edwards Molina et al. 2019), and soybean rust caused by *Phakopsora pachyrhizi* (0.60%) (Dalla Lana et al. 2015). Although defoliation caused by ELS demonstrably reduces peanut yield in Oklahoma, we assume that the lower yield loses experienced by the Spanish market type compared with other pathosystems is a result of the crop defoliation timing, which occurs at the end of the growing season in Oklahoma, and peg strength.

This study provides valuable information to quantify the impact and highlight the importance of ELS as a yield-limiting disease for Spanish market type cultivars in Oklahoma. In summary, there was a significant negative linear relationship between peanut yield and ELS severity in Oklahoma, with intercepts and slopes highly variable across studies. The damage coefficient estimated in our study was determined based on numerous site-years data from fungicide trials

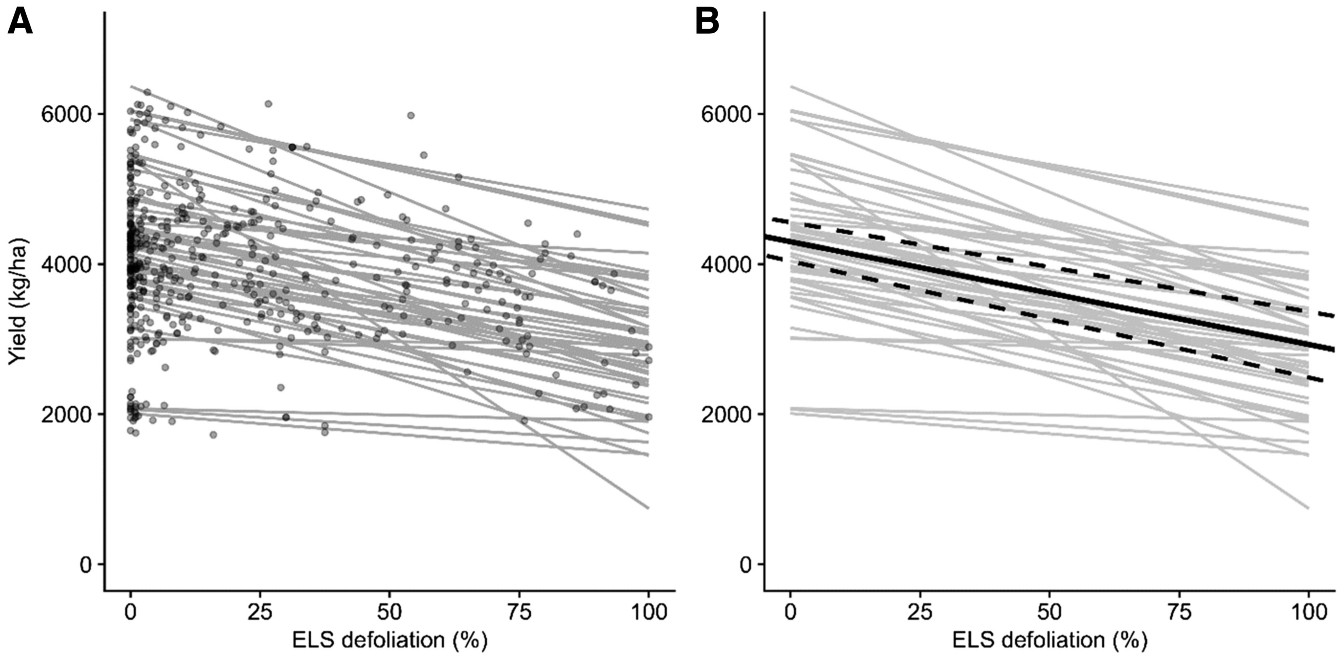

Fig. 2. Results from the fit of the simple linear regression model to peanut yield in relation to early leaf spot (ELS) defoliation (percentage) for 49 studies resulting in study-specific regression lines (gray lines) (**A**) and study-specific prediction lines (gray lines) and population-average predictions (black solid lines) of yield and, respectively, 95% confidence interval (dashed black line) (**B**).

conducted over 26 years under different crop production scenarios and environmental conditions. Our study provides useful generalized information to peanut producers in the Southwestern United States and can help to refine decision-making advisory tools to mitigate ELS in that production region.

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